

2nd

Chapter 7

$$A\theta = y$$

$$A_{n \times d}$$

$$\theta_{d \times 1}$$

$$y_{n \times 1}$$

Design
Matrix

Unknown

Measurement

$$A = \begin{bmatrix} \overline{a_1} \\ \vdots \\ \overline{a_n} \end{bmatrix} \begin{bmatrix} \theta \end{bmatrix} = \begin{bmatrix} \langle a_1, \theta \rangle \\ \vdots \\ \langle a_n, \theta \rangle \end{bmatrix}$$

$$n \gg d$$

Over determined

$$\min_{\theta} \|A\theta - y\|_2^2$$

Under determined
 $n \ll d$

$$A\theta^* = y$$

$$\theta^* + \text{null}(A)$$

$$\text{null}(A) = \{ v \in \mathbb{R}^d \mid Av = 0 \}$$

* Key to make the solution unique, is to make some extra assumption on θ^* .

* Sparsity of θ^*

$$\|\theta^*\|_0 = \# \text{ of non-zero elements of } \theta^*$$

$$\begin{cases} \min_{\theta} \|\theta\|_0 \\ \text{s.t. } A\theta = y \end{cases}$$

$$\left\| \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\|_0 = 3$$

Relaxing the optimization

$$\min_{\theta} \|\theta\|_1$$

$$\text{s.t. } A\theta = y$$

$$\text{Def: } \|\theta\|_1 = \sum_{i=1}^d |\theta_i|$$

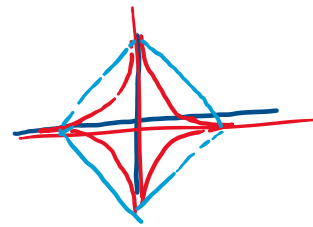
* Can we make sure that ^{the} relaxed problem has the same sparse sol. as the original l_0 -optimization?

l_0 ball

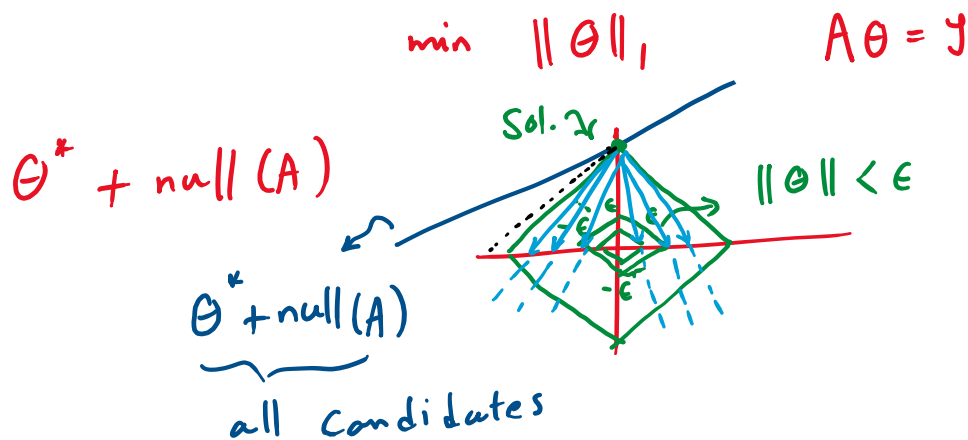
l_p

$p < 1$

$p = 1$



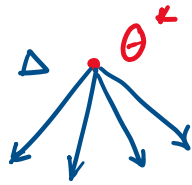
* How to make sure that the sol. to the l_1 -optimization is unique.



* For the uniqueness to happen, we need to make some assumptions on $\text{null}(A)$.

Tangent Cone @ θ^*

$$T(\theta^*) = \{ \Delta \in \mathbb{R}^d \mid \|\theta^* + t\Delta\|_1 \leq \|\theta^*\|_1, \text{ for some } t > 0 \}$$



$\text{Null}(A) \setminus \{0\} \leadsto$ all Δ 's that once added to θ^* , make the new sol. feasible

$$A(\theta^* + \Delta) = y = \underbrace{A\theta^*}_y + \underbrace{A\Delta}_0 = y$$

$$T(\theta^*) \cap (\text{Null}(A) \setminus \{0\}) = \emptyset$$

* Issue: It depends on θ^* that is unknown!

Restricted Null Space Property:

$$\mathcal{L}(S) = \{ \Delta \in \mathbb{R}^d \mid \|\Delta_S\|_1 \geq \|\Delta_{S^c}\|_1 \}$$

Ex. $S = \{1, 3, 5\}$ $S^c = \{2, 4\}$
 $d = 5$

$$\Delta = \begin{bmatrix} -1 \\ 0 \\ 2 \\ 4 \\ 0 \end{bmatrix} \quad S = \{1, 3, 5\} \quad \Delta_S = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} \quad \Delta_{S^c} = \begin{bmatrix} 0 \\ 4 \end{bmatrix}$$

$$\mathcal{L}(S) \cap \text{Null}(A) \setminus \{0\} \stackrel{?}{=} \emptyset$$

Does $\text{null}(A) \setminus \{0\}$ contain a "sparse" vector?

Theorem: The two conditions are equivalent:

- (a) For θ^* with support S , the ℓ_1 -optimization has the unique sol. θ^* , ($A\theta^* = y$).
- (b) A satisfies the restricted null space property.

Pro-: $b \Rightarrow a$ proof by Contradiction:

$\hat{\theta} \neq \theta^*$ that is the sol. ℓ_1 -optimization
 $\hat{\Delta} := \hat{\theta} - \theta^*$. b/c $\hat{\theta}$ is assumed to be the opt. sol.

$$\begin{aligned} \|\theta^*\|_1 &= \cancel{\|\theta_S^*\|_1} \geq \|\hat{\theta}\|_1 \\ &= \|\theta^* + \hat{\Delta}\|_1 \\ &= \|\theta_S^* + \hat{\Delta}_S\|_1 + \|\hat{\Delta}_{S^c}\|_1 \\ &\quad - \|\hat{\Delta}_S\|_1 \\ &\geq \cancel{\|\theta_S^*\|_1} - \|\hat{\Delta}_S\|_1 + \|\hat{\Delta}_{S^c}\|_1 \end{aligned}$$

sub: $|a-b| \geq |a| - |b|$

$$\Rightarrow \|\hat{\Delta}_S\|_1 \geq \|\hat{\Delta}_{S^c}\|_1$$

$$\hat{\Delta} \stackrel{?}{\in} \text{null}(A)$$

$$\begin{aligned} A\theta^* &= y \\ A\hat{\theta} &= y \end{aligned} \Rightarrow A\hat{\Delta} = 0 \rightarrow \hat{\Delta} \in \text{null}(A)$$

NSP assumes that no such $\hat{\Delta}$ does not exist.

✗

$a \Rightarrow b$

Proof:

$\theta^* \in \text{null}(A)$

$$\min_{\theta} \|\theta\|, \quad \text{s.t. } A\theta = A \underbrace{\begin{bmatrix} \theta_s^* \\ \underline{0} \end{bmatrix}}_y$$

is the unique sol. b/c
it satisfies $A\theta = y$, & $\text{supp}(\begin{bmatrix} \theta_s^* \\ \underline{0} \end{bmatrix}) = S$.

$\begin{bmatrix} 0 \\ -\theta_{s^c}^* \end{bmatrix}$ is also a sol.
Candidate

$$A\theta^* = \underline{0} \quad A \begin{bmatrix} \theta_s^* \\ \theta_{s^c}^* \end{bmatrix} = \underline{0}$$

$$\Rightarrow A \left(\begin{bmatrix} \theta_s^* \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ \theta_{s^c}^* \end{bmatrix} \right) = \underline{0}$$

$$\Rightarrow A \begin{bmatrix} \theta_s^* \\ 0 \end{bmatrix} = -A \begin{bmatrix} 0 \\ \theta_{s^c}^* \end{bmatrix} = A \begin{bmatrix} 0 \\ -\theta_{s^c}^* \end{bmatrix}$$

$\rightarrow$ b/c unique sol. assumed in a,

$$\left\| \begin{bmatrix} 0 \\ -\theta_{s^c}^* \end{bmatrix} \right\|_1 > \left\| \begin{bmatrix} \theta_s^* \\ 0 \end{bmatrix} \right\|_1$$

$$\Rightarrow \|\theta_{s^c}^*\|_1 > \|\theta_s^*\|_1$$

$\Rightarrow$ null space prop. is satisfied

Mutual Incoherence Property

$$A = \begin{bmatrix} A_1 & A_2 & \dots & A_d \end{bmatrix}$$

$$\max_{i,j} \left| \frac{\langle A_i, A_j \rangle}{n} - \delta_{ij} \right| \leq \frac{1}{3} s \quad \downarrow \quad \|\theta^*\|.$$

$\Rightarrow$ A satisfies the NSP(s).